3.4. ISOTOPIC RATIOS IN NATURE

Mass spectrometric investigations of geologic material has shown that isotopic ratios vary slightly in nature with the largest variations observed for the heaviest and lightest elements. Hence the atomic weight of many elements is a little uncertain, which is indicated be giving fewer decimals than for their isotopes.

For the heaviest elements the reason for the variation in the ratio of isotopes in an element can be related directly to the consequence of naturally occurring radioactive decay. For example, in a thorium mineral the decay sequence terminates in the formation of an isotope of lead with mass number 208. By contrast, in a uranium ore, in which ^{238}U is the most abundant uranium isotope, the primary end product of the decay is an isotope of lead with mass number 206 (see Fig. 1.2). This means that thorium minerals and uranium minerals contain lead with different isotopic composition. In fact, one of the best confirmations of the existence of the radioactive decay series came from the proof that lead had different net atomic weights in different minerals.

Lead from different ores will contain slightly different isotopic abundances due to the fact that all long natural radioactive decay series end up in differet isotopes of lead An unusual use of this fact was made by Andrasko et al. to identify smears and fragments from lead bullets used in a homicide case, so that the suspect could be bound to the case, as the isotopic composition of lead bullets can be identified not only by manufacturer but also by manufacturing date.

The isotopic ratios for the lightest elements depend on in which natural material they are found. The ratio for $^{7}Li/^{6}Li$ varies from 12.20 to 12.48, while that of $^{12}C/^{13}C$ varies from 89.3 to 93.5. The isotopic ratio $^{18}O/^{16}O$ varies by almost 5% as shown in Fig. 3.3. However, since natural oxygen contains only 0.2% ^{18}O, even a 5% variation in the isotopic ratio has very little influence on the atomic weight of oxygen. The variation of natural isotopic ratios for boron and chlorine causes the uncertainties in their abundances shown in Table 3.

3.5. PHYSICOCHEMICAL DIFFERENCES FOR ISOTOPES[2]

Although the isotopic variations in the heaviest elements can be attributed to the consequences of radioactive decay, the variations observed in lighter elements are attributable to chemical behaviour. The rates and equilibria of chemical reactions depend on the masses of the atoms involved in the reactions as is explained in §3.6 and §3.7. As a consequence, isotopes may be expected to have somewhat different quantitative physicochemical values (the *isotope effect*). This effect is naturally more predominant for lighter elements than heavier since for the lighter elements the change in mass by adding a neutron

[2] §§3.5-3.7 outline the scientific basis for isotope effects and isotope separation. This is not essential for the Radiochemistry part of this book.